Symmetric Matrices & The Spectral Theorem

Symmetric Matrices

Recall that if $A \in Mat_n(\mathbb{F})$ is symmetric, then $A^T = A$.

Theorem. Let $A \in Mat_n(\mathbb{R})$ be symmetric, then all eigenvalues of A are real.

Eigenvectors of Symmetric Matrices

Theorem. Let $A \in Mat_n(\mathbb{R})$ be symmetric, let λ, μ be two different eigenvalues of A, with eigenvectors $\vec{v}, \vec{w}$ respectively. Then $\vec{v}$ and $\vec{w}$ are orthogonal, i.e. $\vec{v} \cdot \vec{w} = 0$.

Proof. If $A\vec{v} = \lambda\vec{v}$, $A\vec{w} = \mu\vec{w}$, then

$$\begin{aligned}
\lambda(\vec{v} \cdot \vec{w}) &= (\lambda\vec{v}) \cdot \vec{w} = (A\vec{v}) \cdot \vec{w} \\
&= (A\vec{v})^T \vec{w} \\
&= \vec{v}^T A^T \vec{w} \\
&= \vec{v}^T A \vec{w} \quad (A \text{ is symmetric, } A = A^T). \\
&= \vec{v}^T (\mu\vec{w}) = \mu(\vec{v} \cdot \vec{w}).
\end{aligned}$$

Thus, $\lambda(\vec{v}\cdot\vec{w}) = \mu(\vec{v}\cdot\vec{w})$, so

$$(\lambda-\mu)(\vec{v}\cdot\vec{w}) = 0.$$

But $\lambda \neq \mu$, so $\lambda - \mu \neq 0 \Rightarrow \vec{v}\cdot\vec{w} = 0$.

<u>Theorem (Spectral Theorem for Symmetric Matrices).</u>

Let $A \in Mat_n(\mathbb{R})$ be a symmetric matrix, then

① A is diagonalisable;

② we can find a matrix P and D such that

$$A = PDP^{-1}$$

where the columns of P are orthonormal.

Property ② is known as "orthogonally diagonalisable".

Any real symmetric matrix is diagonalisable. So, there is a basis of $\mathbb{R}^n$ of eigenvectors of A.

Choosing P to have orthonormal columns, we get an orthonormal basis of $\mathbb{R}^n$.

Theorem. Let P be any matrix with orthonormal columns. Then $P^{-1} = P^T$.

Proof. Write

$$P = \begin{pmatrix} | & & | \\ \vec{p}_1 & \cdots & \vec{p}_n \\ | & & | \end{pmatrix}, \quad P^T = \begin{pmatrix} - & \vec{p}_1^{\,T} & - \\ & \vdots & \\ - & \vec{p}_n^{\,T} & - \end{pmatrix}.$$

$$P^T P = \begin{pmatrix} \|\vec{p}_1\|^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\vec{p}_n\|^2 \end{pmatrix} = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix} = I_n$$

$$\therefore \; P^T = P^{-1}.$$

Corollary. Let A be a real symmetric matrix, then A can be written as

$$A = PDP^{-1}$$

for a diagonal matrix D and matrix P with orthonormal columns.

Introduction to Jordan Canonical Forms

Definition (Jordan Block). A Jordan block $J_{n,\lambda}$ for a scalar λ is an $n \times n$ upper triangular matrix of the form

$$J_{n,\lambda} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix}$$

Examples.

$$J_{4,3} = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

$$J_{2,0} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Definition (Block diagonal matrix). A block diagonal matrix is a square matrix of the form

$$M = \begin{pmatrix} M_1 & & & 0 \\ & M_2 & & \\ & & \ddots & \\ 0 & & & M_n \end{pmatrix}$$

where each M_i is a square matrix.

<u>Example</u>

$$M = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 1 & 0 & 9 \end{pmatrix} \qquad M_1 = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \qquad M_2 = (7) \qquad M_3 = \begin{pmatrix} 1 & 3 & 4 \\ 2 & -1 & 3 \\ 1 & 0 & 9 \end{pmatrix}$$

<u>Theorem (Jordan Canonical Forms).</u>

Let $A \in \mathrm{Mat}_n(\mathbb{C})$. Then there are matrices P, J such that

$$A = PJP^{-1}$$

and J is a block diagonal matrix of Jordan blocks $J_{k,\lambda}$ corresponding to eigenvalues λ of A.

We call J the Jordan Canonical Form of A.

$$\boxed{J = \begin{pmatrix} \lambda_1 & 1 & & & & & & & & & & \\ & \lambda_1 & 1 & & & & & & & & & \\ & & \lambda_1 & \ddots & & & & & & & & \\ & & & \lambda_1 & & & & & & & & \\ & & & & \lambda_2 & 1 & & & & & & \\ & & & & & \lambda_2 & 1 & & & & & \\ & & & & & & \lambda_2 & \ddots & & & & \\ & & & & & & & \lambda_2 & & & & \\ & & & & & & & & \lambda_k & 1 & & \\ & & & & & & & & & \lambda_k & \ddots & \\ & & & & & & & & & & \ddots & 1 \\ & & & & & & & & & & & \lambda_k \end{pmatrix}}$$

Any diagonal matrix is in JCF.

$$\begin{pmatrix} \boxed{\lambda_1} & & \\ & \ddots & \\ & & \boxed{\lambda_k} \end{pmatrix}$$

<u>Example</u>.

$$\begin{pmatrix} \boxed{\begin{matrix} 2 & 1 \\ 0 & 2 \end{matrix}} & & \\ & \boxed{\begin{matrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}} & \\ & & \boxed{7} \end{pmatrix}$$

is in JCF.